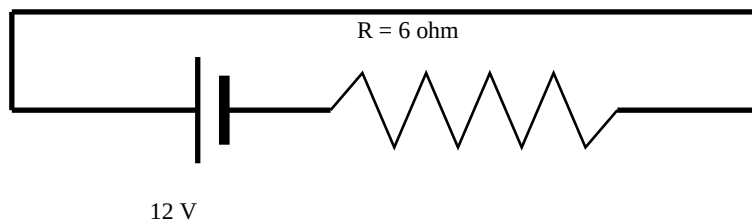


Q1. A particle starts from rest and reaches a speed of 24 m/s in 6 s with uniform acceleration. What is its acceleration?

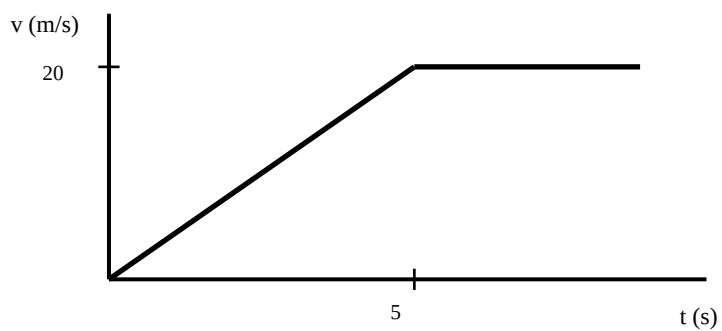
- A. 2 m/s^2
- B. 4 m/s^2
- C. 6 m/s^2
- D. 8 m/s^2

Q2. A 12 V battery is connected to a 6 ohm resistor as shown. What current flows in the circuit?



- A. 0.5 A
- B. 1 A
- C. 2 A
- D. 72 A

Q3. The velocity-time graph shown is for a body starting from rest. What is its acceleration during the first 5 s?

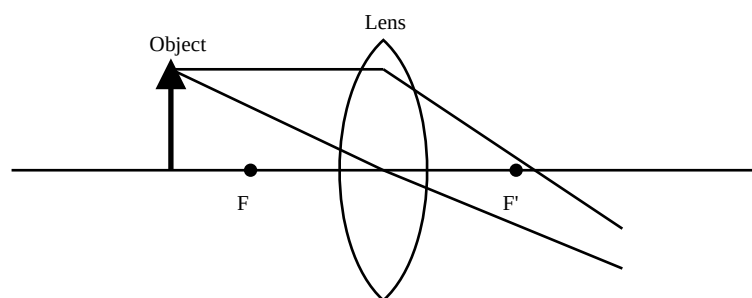


- A. 2 m/s^2
- B. 4 m/s^2
- C. 5 m/s^2
- D. 20 m/s^2

Q4. A convex lens has focal length 20 cm. Its power is:

- A. +2 D
- B. +5 D
- C. -5 D
- D. +0.2 D

Q5. In the ray diagram shown, the object is placed between F and $2F$ of a convex lens. The image formed is generally:



- A. Virtual, erect, diminished
- B. Real, inverted, enlarged
- C. Real, erect, same size
- D. Virtual, inverted, enlarged

Q6. Two resistors of 4 ohm and 12 ohm are connected in parallel. Their equivalent resistance is:

- A. 3 ohm
- B. 4 ohm
- C. 8 ohm
- D. 16 ohm

Q7. A wave has frequency 50 Hz and wavelength 2 m. Its speed is:

- A. 25 m/s
- B. 50 m/s
- C. 100 m/s
- D. 200 m/s

Q8. An electron enters a uniform magnetic field perpendicular to the field lines. Its path is:

- A. Straight line
- B. Circular
- C. Parabolic
- D. Random

Q9. A 2 kg body is raised vertically through 5 m. Taking $g = 10 \text{ m/s}^2$, the gain in potential energy is:

- A. 10 J
- B. 25 J
- C. 50 J
- D. 100 J

Q10. If the frequency of incident light is below the threshold frequency of a metal, increasing intensity will:

- A. Eject more electrons
- B. Increase electron kinetic energy
- C. Not cause photoelectric emission
- D. Reduce work function